

Mutually Independent Hamiltonian Cycles of Double Loop Graphs

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Abstract

Let $G=(V,E)$ be a graph of order n . A Hamiltonian cycle of G is a cycle that contains every vertex in G . Two Hamiltonian cycles $C_1=\langle u_1, u_2, \dots, u_n, u_1 \rangle$ and $C_2=\langle v_1, v_2, \dots, v_n, v_1 \rangle$ of G are independent if $u_1=v_1$ and $u_i \neq v_i$ for $2 \leq i \leq n$. A set of Hamiltonian cycles $\{C_1, C_2, \dots, C_k\}$ of G is mutually independent if its elements are pairwise independent. The mutually independent hamiltonicity $IHC(G)$ of a graph G is the maximum integer k such that for any vertex u of G there are k mutually independent Hamiltonian cycles of G starting at u . For double loop network graphs $DL(n; a, b)$, this paper showed that $IHC(DL(n; 1, 2))=4$ for $n \geq 7$, and $IHC(DL(n; 1, 3))=4$ for odd $n \geq 7$ and even $n \geq 12$.

1 Introduction

The interconnection network, a crucial step on designing a massively multiprocessor system, is an important application of graph theory. An interconnection network can be represented by a graph whose vertices represent processors and edges represent the connections between two vertices.

Let $G=(V,E)$ be a graph with vertex set V and edge set E . A Hamiltonian path of a graph is a path that traverses all vertices in G once. A Hamiltonian cycle of graph G is a cycle that visits all vertices in G once. Without considering the edge bandwidth of G , one may be able to find some Hamiltonian cycles, which have the same start vertex but the cycles do not visit the same vertex on the same step. Two Hamiltonian cycles $C_1=\langle v_1, v_2, \dots, v_n, v_1 \rangle$ and $C_2=\langle u_1, u_2, \dots, u_n, u_1 \rangle$ are independent in G if $v_1=u_1$ and $v_i \neq u_i$ for $2 \leq i \leq n$. A set of Hamiltonian cycles $\{C_1, C_2, \dots, C_k\}$ of G is mutually independent if its

elements are pairwise independent. The Maximum number of mutually independent Hamiltonian cycles in graph G was denoted by $IHC(G)$. Because there are many mutually conflicting requirements in designing the topology for computer network, there are many studies on mutually independent Hamiltonian cycles of graphs, especially regular graphs, including hypercube[2][6][16][17], pancake graph[9][10], star[1][8][10], k -ary n -cube graph[4][11][18][19], wrapped butterfly[5], bubble-sort graph[15], dual cubes[14], and burn pancake graph[7][20].

The Double-Loop graphs, a special case of multiloop graphs, are extensions of ring graphs. Because of the shorter delay, higher reliability and lower-cost hardware, double-loop structure is widely used in the design and implementation of local network and parallel processing architectures. A double loop network $DL(n; a, b)$ is a directed graph with n vertices $\{0, 1, 2, \dots, n-1\}$ and $2n$ directed edges such that vertex i is adjacent to vertex $i+a \pmod{n}$ and $i+b \pmod{n}$, refers to a -links and b -links respectively. Here we consider the undirected version of the double loop network. For special value of a and b , $DL(n; 1, n-1)$ was first proposed by Liu[11], named as *distributed double-loop computer network*, and the class $DL(n; 1, b)$ for $2 \leq b \leq n-1$ was proposed by Raghavendra et al.[13]. Figure 1 shows an example of $G=DL(8; 1, 3)$ and $G=DL(8; 1, 2)$. The $IHC(DL(n; 1, b))$ was studied in [20] which only discussed the case of odd n and requested $\gcd(n, b)=1$ to have $IHC(G)=4$. This paper shows that $IHC(DL(n; 1, 2))=4$ for $n \geq 7$, and $IHC(DL(n; 1, 3))=4$ for odd $n \geq 7$ and even $n \geq 12$. We also show that $DL(6; 1, 2)$, $DL(8; 1, 3)$ and $DL(10; 1, 3)$, each of them only has three mutually independent Hamiltonian cycles.

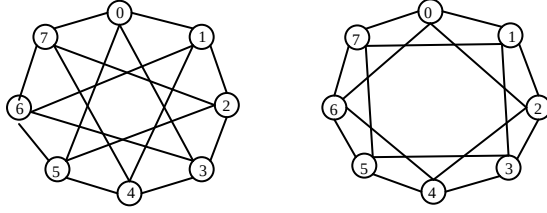


Fig 1. $G = DL(8; 1, 3)$ and $G = DL(8; 1, 2)$

2 Main Result

Since $DL(5; 1, 2) \cong K_5$, and it's clear that $IHC(K_n) = n - 1$, hence we know that $IHC(DL(5; 1, 2)) = 4$.

Theorem 1: $IHC(DL(6; 1, 2)) = 3$.

Proof: Since every vertex is similar in double loop network, without loss of generality, let vertex 0 be the origin vertex. Since vertex 3 is not adjacent to vertex 0, starting from vertex 0, vertex 3 cannot be put in the first and last position in the cycle. Then there is only three positions for vertex 3, hence $IHC(DL(6; 1, 2)) \leq 3$. Now consider $C_1 = \langle 0, 1, 2, 3, 4, 5, 6, 0 \rangle$, $C_2 = \langle 0, 2, 3, 4, 5, 6, 1, 0 \rangle$ and $C_3 = \langle 0, 6, 1, 2, 3, 4, 5, 0 \rangle$. Clearly, C_1, C_2, C_3 are mutually independent in $DL(6; 1, 2)$, that is $IHC(DL(6; 1, 2)) = 3$. ■

Since every vertex is similar in double loop networks, without loss of generality, we may let vertex 0 be the origin vertex when considering the Hamiltonian cycles in double loop networks.

Theorem 2: $IHC(DL(n; 1, 2)) = 4$, for $n \geq 7$.

Proof: We construct four Hamiltonian cycles of $DL(n; 1, 2)$ as follows.

First, go through all the 1-links from vertex 1 to $n - 1$, we have $C_1 = \langle 0, 1, \dots, n - 1, 0 \rangle$. Next, after a 2-link to vertex 2, continues to go through all the 1-links to $n - 1$, then by a 2-link to vertex 1 and return to vertex 0, then $C_2 = \langle 0, 2, 3, 4, \dots, n - 1, 1, 0 \rangle$. Third, use a 1-link to vertex $n - 1$, then a 2-link to vertex 1 and continues to go through all the 1-links to $n - 2$, then a 2-link back to the origin, so we have $C_3 = \langle 0, n - 1, 1, 2, \dots, n - 2, 0 \rangle$. Notice that these three Hamiltonian cycles are a shift from each other except the in and out portion and are clearly mutually independent. For the last cycle, use 2-links to walk from $n - 2$ to half of the vertices

then back, after finish the second half of the vertices, then continue with the first half, depend on the number of n and $\lceil n/2 \rceil$, we have the fourth cycle as follows:

Case 1: n and $\lceil n/2 \rceil$ are both even or both odd. Then $C_4 = \langle 0, n - 2, n - 4, \dots, \lceil n/2 \rceil, \lceil n/2 \rceil + 1, \lceil n/2 \rceil + 3, \dots, n - 1, 1, 3, \dots, \lceil n/2 \rceil - 1, \lceil n/2 \rceil - 2, \lceil n/2 \rceil - 4, \dots, 2, 0 \rangle$.

Case 2: one of n and $\lceil n/2 \rceil$ is even and the other is odd. Then $C_4 = \langle 0, n - 2, n - 4, \dots, \lceil n/2 \rceil + 1, \lceil n/2 \rceil, \lceil n/2 \rceil + 2, \dots, n - 1, 1, 3, \dots, \lceil n/2 \rceil - 1, \lceil n/2 \rceil - 2, \lceil n/2 \rceil - 4, \dots, 2, 0 \rangle$.

Notice that when C_4 is walking through the second half of the vertices, C_1, C_2 , and C_3 are in the first half of the vertices and vice versa. Hence C_1, C_2, C_3 , and C_4 are mutually independent Hamiltonian cycles of $DL(n; 1, 2)$. Since $DL(n; 1, 2)$ is 4-regular, we conclude that $IHC(DL(n; 1, 2)) = 4$ for $n \geq 7$. ■

To simplify the terms, from now on, in this paper, we call those vertices with even labels as *even vertices* and those vertices with odd label as *odd vertices*.

Theorem 3: Let $G = DL(n; 1, 3)$ for $n = 8$ or $n = 10$. Then $IHC(G) = 3$.

Proof: As described in Theorem 1, without of generality, let vertex 0 be the starting vertex. Notice that, in $G = DL(n; 1, 3)$ with even n , every odd vertex is only adjacent to the even vertices and every even vertex is only adjacent to odd vertices. When $n = 8$, since there are only three even vertices in $DL(8; 1, 3)$ besides vertex 0, for each even step position in the Hamiltonian cycle starting from vertex 0 can only be occupied by three even vertices. Hence there is no more than three mutually independent Hamiltonian cycles in $DL(8; 1, 3)$ which implies $IHC(DL(8; 1, 3)) \leq 3$. Now consider $C_1 = \langle 0, 1, 2, 3, 4, 5, 6, 7, 0 \rangle$, $C_2 = \langle 0, 3, 4, 5, 6, 7, 2, 1, 0 \rangle$ and $C_3 = \langle 0, 7, 6, 1, 2, 3, 4, 5, 0 \rangle$. Then $\{C_1, C_2, C_3\}$ is mutually independent. That is, $IHC(DL(8; 1, 3)) = 3$. For $n = 10$, since vertex 5 is not adjacent to vertex 0, similar to Theorem 1, there is only three positions for vertex 5. Hence we have $IHC(DL(10; 1, 3)) \leq 3$. Let $C'_1 = \langle 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 0 \rangle$, $C'_2 = \langle 0, 3, 4, 5, 6, 7, 8, 9, 2, 1, 0 \rangle$ and $C'_3 = \langle 0, 9, 8, 1, 2, 3, 4, 5, 6, 7, 0 \rangle$.

Then C'_1, C'_2, C'_3 are three mutually independent Hamiltonian cycles in $DL(10;1,3)$. That is $IHC(DL(n;1,3)) = 3$ for $n \in \{8, 10\}$. ■

Theorem 4: Let $G = DL(n;1,3)$ for odd $n \geq 7$. Then $IHC(G) = 4$.

Proof: We construct four Hamiltonian cycles of $DL(n;1,3)$ as follows.

First, go through all the 1-links from vertex 1 to $n-1$, we have $C_1 = \langle 0, 1, \dots, n-1, 0 \rangle$. Next, after a 3-link to vertex 3, continues to go through all the 1-links to $n-1$, then by a 3-link to vertex 2 and a 1-link to vertex 1 then return to vertex 0, then $C_2 = \langle 0, 3, 4, \dots, n-1, 2, 1, 0 \rangle$. Third, use two 1-links to vertex $n-2$, then a 2-link to vertex 1 and continues to go through all the 1-links to $n-3$, then a 3-link back to the origin, so we have $C_3 = \langle 0, n-1, n-2, 1, 2, \dots, n-3, 0 \rangle$. Notice that these three Hamiltonian cycles are a shift from each other except the in and out portion and are clearly mutually independent.

For the fourth cycle, from vertex 0, we use a 3-link to vertex $n-3$, follow by a 1-link to $n-4$; then use a 3-link to vertex $n-1$ follow by a 1-link to $n-2$ and a 3-link to $n-5$, then use 1-links go all the way to vertex 5, follow by a 3-link to vertex 2, then a 1-links to vertex 1 follow by a 3-link to vertex 4 then a 1-link to vertex 3 and back to vertex 0. Hence $C_4 = \langle 0, n-3, n-4, n-1, n-2, n-5, n-6, \dots, 5, 2, 1, 4, 3, 0 \rangle$. Since C_4 is at vertex with even labels for each odd step, and at vertex with odd labels for each even step, which is opposite of C_1, C_2 (except for the last two steps), and C_3 (except for the first two steps), hence, $\{C_1, C_2, C_3, C_4\}$ are mutually independent Hamiltonian cycles of $DL(n;1,3)$ for $n \geq 11$. For $n=7$, let $C_4 = \langle 0, 4, 1, 2, 5, 6, 3, 0 \rangle$, then $\{C_1, C_2, C_3, C_4\}$ are mutually independent Hamiltonian cycles of $DL(7;1,3)$. For $n=9$, consider $C'_3 = \langle 0, 8, 5, 4, 7, 1, 2, 3, 6, 0 \rangle$ and $C_4 = \langle 0, 6, 7, 8, 5, 2, 1, 4, 3, 0 \rangle$, then $\{C_1, C_2, C'_3, C_4\}$ are mutually independent Hamiltonian cycles of $DL(9;1,3)$. That is, $IHC(DL(n;1,3)) = 4$ for $n \geq 7$ when n is odd. ■

Theorem 5: For $G = DL(n;1,3)$ for even $n \geq 12$, $IHC(G) = 4$.

Proof: Let C_1, C_2, C_3 be three Hamiltonian cycles of $DL(n;1,3)$ described as in the proof of

Theorem 4. For the fourth cycle, the idea is using 3-links to go from vertex $n-3$ to near vertex $n/2$, then by using a 1-link after a 3-link alternatively to cover all the vertices from $n/2$ to $n-1$. After that, we use a 3-link to vertex 2, then use a 1-link after a 3-link alternatively to go from vertex 1 to the first half of the vertices while skipping all the vertices $3t$ for $1 \leq t \leq k-1$. Finally, use 3-links to go all the way back to vertex 3 then vertex 0. In this way, when $n \geq 14$, we have C_4 going through latter half of the vertices first while C_1, C_2 , and C_3 are in the first half and vice versa. Hence $\{C_1, C_2, C_3, C_4\}$ is mutually independent of $DL(n;1,3)$ for even $n \geq 14$. To be more precisely, when $n=6k$ for even $k \geq 4$, $C_4 = \langle 0, n-3, n-6, \dots, 3k, 3k-1, \dots, n-1, 2, 1, 4, \dots, 3k-2, 3k-3, 3k-6, \dots, 3, 0 \rangle$; and for odd $k \geq 3$, $C_4 = \langle 0, n-3, n-6, \dots, 3k, 3k+1, 3k+2, 3k+5, \dots, n-1, 2, 1, 4, \dots, 3k-1, 3k-2, 3k-3, 3k-6, \dots, 3, 0 \rangle$. When $n=6k+2$ for even $k \geq 2$, $C_4 = \langle 0, n-3, n-6, \dots, 3k+2, 3k+1, 3k+4, L, n-1, 2, 1, 4, L, 3k-2, 3k-1, 3k, 3k-3, L, 3, 0 \rangle$; and for odd $k \geq 3$, $C_4 = \langle 0, n-3, \dots, 3k+2, 3k+3, \dots, n-1, 2, 1, \dots, 3k+1, 3k, \dots, 3, 0 \rangle$. When $n=6k+4$ for even $k \geq 2$, $C_4 = \langle 0, n-3, \dots, 3k+4, 3k+3, 3k+6, \dots, n-1, 2, 1, \dots, 3k+2, 3k+1, 3k, \dots, 3, 0 \rangle$; and for odd $k \geq 3$, $C_4 = \langle 0, n-3, n-6, \dots, 3k+4, 3k+5, 3k+6, \dots, n-1, 2, 1, \dots, 3k+1, 3k+2, 3k+3, 3k, \dots, 3, 0 \rangle$. For $n=12$, let $C_4 = \langle 0, 9, 6, 7, 8, 11, 10, 1, 2, 5, 4, 3, 0 \rangle$. Then in all the cases, $\{C_1, C_2, C_3, C_4\}$ is mutually independent of $DL(n;1,3)$ for even $n \geq 12$. That is $IHC(G) = 4$. ■

3 Conclusion

Double loop is an interconnection network whose underlining structure is as simple as ring but with higher connectivity and shorter diameter. In double loop network structure, $DL(n;1,2)$ and $DL(n;1,3)$ get more attention then others. This paper showed that with enough nodes, a double-loop can find four mutually independent Hamiltonian cycles. The hamiltonicity $IHC(DL(n;1,2)) = 4$ for $n \geq 7$; and $IHC(DL(n;1,3)) = 4$ for odd $n \geq 7$ and even $n \geq 12$ are proved. We also showed that

$IHC(DL(6; 1, 2)) = 3$ and $IHC(DL(n; 1, 3)) = 3$ for $n \in \{8, 10\}$.

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